BIO5105 – CELL BIOLOGY

FINAL TERM IMPORTANT QUESTIONS

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1. What is the isoaccepting tRNA?

Isoaccepting tRNAs are different tRNA molecules that accept and carry the same amino acid but possess different anticodon sequences. Because of their different anticodons, they recognize different codons that specify the same amino acid according to the genetic code. This allows a single amino acid to be incorporated into a growing polypeptide chain in response to multiple synonymous codons. Isoaccepting tRNAs play an important role in maintaining the efficiency and accuracy of protein synthesis and help ensure the proper translation of genetic information into proteins.

2. Differentiate between Eukaryotic and Prokaryotic RNA.

Difference between Eukaryotic RNA and Prokaryotic RNA

Eukaryotic RNA	Prokaryotic RNA
Synthesized in the nucleus and functions mainly in the cytoplasm.	Synthesized and functions in the cytoplasm because there is no nucleus.
mRNA is processed by 5′ capping, 3′ poly(A) tail addition, and splicing.	mRNA is not processed; it lacks a 5′ cap, poly(A) tail, and intron splicing.
mRNA is usually monocistronic, coding for one protein.	mRNA is usually polycistronic, coding for multiple proteins.
Contains three main RNA polymerases (RNA polymerase I, II, and III).	Contains one RNA polymerase that synthesizes all types of RNA.
Introns are commonly present in genes.	Introns are generally absent in genes.
Transcription and translation occur separately.	Transcription and translation occur simultaneously.

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3. Write a short note on mutsβ complex.

MutSβ Complex

The MutSβ complex is a DNA mismatch repair (MMR) protein complex found in eukaryotic cells. It is composed of two proteins, MSH2 and MSH3. Its main function is to recognize and bind insertion/deletion loops (IDLs) that are formed during DNA replication, especially loops containing 2–13 unpaired nucleotides**.** After recognizing the error, MutSβ recruits other mismatch repair proteins, such as MutL complexes, to remove the incorrect DNA segment and synthesize the correct sequence. This repair process maintains genome stability, reduces mutations, and helps prevent diseases such as cancer caused by defects in DNA repair.

4. What happens when DNA fails to replicate?

When DNA fails to replicate**,** the cell cannot produce an exact copy of its genetic material, making normal cell division impossible. This activates DNA damage and replication checkpoints, which temporarily stop the cell cycle to allow repair of the damaged DNA. If replication cannot be completed successfully, the cell may undergo apoptosis (programmed cell death) or enter cellular senescence, where it permanently stops dividing. Persistent failure of DNA replication can lead to genomic instability, chromosome breakage**,** mutations, and defective daughter cells. In multicellular organisms, these abnormalities may contribute to genetic disorders, tissue dysfunction, aging, and the development of cancer.

5. Differentiate between reciprocal translocation and ribosomal translocation.

Difference between Reciprocal Translocation and Ribosomal Translocation

Reciprocal Translocation	Ribosomal Translocation
A chromosomal mutation in which segments are exchanged between two non-homologous chromosomes**.**	A step of protein synthesis in which the ribosome moves one codon forward along the mRNA.
Occurs during DNA/chromosome rearrangement.	Occurs during translation on the ribosome.
May disrupt genes or alter gene expression, leading to genetic disorders or cancer.	Ensures the correct addition of amino acids to the growing polypeptide chain.
Involves the exchange of chromosome fragments.	Involves the movement of mRNA and tRNAs through the ribosome.
Does not require ribosomes or translation factors.	Requires elongation factors (EF-G in prokaryotes/eEF2 in eukaryotes) and GTP for movement.

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6. Differentiate between anaphase I and anaphase II meiosis.

Difference between Anaphase I and Anaphase II

Anaphase I	Anaphase II
Occurs during Meiosis I.	Occurs during Meiosis II.
Homologous chromosomes separate and move to opposite poles.	Sister chromatids separate and move to opposite poles.
Centromeres do not divide.	Centromeres divide, allowing sister chromatids to separate.
The chromosome number is reduced from diploid (2n) to haploid (n).	The chromosome number remains haploid (n).
Each chromosome still consists of two sister chromatids.	Each separated chromosome consists of a single chromatid.
Responsible for reductional division.	Responsible for equational division.

7. How the transcription is initiated in eukaryotes?

Transcription in eukaryotes is initiated when general transcription factors (GTFs) bind to the promoter region of a gene, usually at the TATA box. The TATA-binding protein (TBP), a component of TFIID, first recognizes and binds to the promoter. Other transcription factors then assemble with RNA polymerase II to form the pre-initiation complex. Next, TFIIH unwinds the DNA double helix and phosphorylates the C-terminal domain (CTD) of RNA polymerase II. This activates RNA polymerase II, allowing it to move along the DNA template and begin RNA synthesis in the 5′ → 3′ direction. The process is further regulated by activators, enhancers, and mediator proteins, which ensure accurate and efficient transcription.

8. Define mutS and mutL.

MutS:

MutS is a DNA mismatch repair (MMR) protein that recognizes and binds to mismatched base pairs and small insertion/deletion loops formed during DNA replication. It initiates the mismatch repair process by detecting replication errors and recruiting other repair proteins.

MutL:

MutL is a DNA mismatch repair protein that interacts with MutS after a mismatch is recognized. It acts as a mediator, coordinating the repair process by activating downstream repair proteins, including MutH in bacteria, to remove the incorrect DNA segment and restore the correct DNA sequence.

9. What are microfilaments?

Microfilaments

Microfilaments are thin, solid protein filaments of the cytoskeleton composed mainly of actin. They are about 7 nm in diameter and are found throughout the cytoplasm of eukaryotic cells, especially beneath the plasma membrane. Microfilaments provide mechanical support and help maintain the shape of the cell. They are involved in cell movement, muscle contraction, cytokinesis (cell division), cytoplasmic streaming, and the formation of structures such as microvilli. Because they are highly dynamic, microfilaments continuously assemble and disassemble to meet the changing needs of the cell.

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10. What are spliceosomes? Write its examples.

Spliceosomes

Spliceosomes are large ribonucleoprotein (RNP) complexes present in the nucleus of eukaryotic cells. Their main function is to remove introns from pre-mRNA and join exons together during RNA splicing, producing mature mRNA that can be translated into protein. A spliceosome is made up of small nuclear RNAs (snRNAs) and proteins, collectively called small nuclear ribonucleoproteins (snRNPs)**.** ***Examples:*** The major snRNPs that form the spliceosome are: U1 snRNP, U2 snRNP, U4 snRNP, U5 snRNP & U6 snRNP.

11. What is cyclin-CDK complex in DNA replication?

The Cyclin-CDK (Cyclin-dependent kinase) complex is a group of regulatory proteins that controls the progression of the cell cycle and ensures that DNA replication occurs only once during each cell cycle. Cyclins bind to and activate CDKs, forming an active complex. During the G1 phase**,** the Cyclin D–CDK4/6 and Cyclin E–CDK2 complexes prepare the cell for DNA replication by promoting the transition from G1 to S phase. In the S phase**,** the Cyclin A–CDK2 complex initiates and regulates DNA replication by activating replication origins and preventing re-replication of DNA. Thus, the Cyclin-CDK complex ensures accurate DNA replication, proper cell cycle progression, and maintenance of genomic stability.

12. Explain principle of independent of alleles of different genes.

The principle of independent assortment, proposed by Gregor Mendel, states that the alleles of different genes assort independently of one another during gamete formation. This means that the inheritance of one pair of alleles does not affect the inheritance of another pair, provided the genes are located on different chromosomes or are far apart on the same chromosome. During metaphase I of meiosis, homologous chromosome pairs align randomly, and in anaphase I they separate independently. As a result, different combinations of maternal and paternal chromosomes are distributed into gametes, producing genetic variation in the offspring. This principle explains why new combinations of traits appear in each generation.

13. Differentiate between sex chromosomes and autosomes.

Difference between Sex Chromosomes and Autosomes

Sex Chromosomes	Autosomes
Determine the sex of an individual.	Control body traits other than sex.
Humans have 1 pair of sex chromosomes.	Humans have 22 pairs of autosomes.
Represented as X and Y chromosomes.	Numbered from 1 to 22**.**
Females have XX and males have XY chromosomes.	The same in both males and females.
Carry genes related to sex determination and sex-linked traits.	Carry genes responsible for most physical and physiological characteristics**.**

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14. Write a short note on trisomy.

Trisomy

Trisomy is a chromosomal abnormality in which an individual has three copies of a chromosome instead of the normal two. It usually occurs due to nondisjunction during meiosis, when homologous chromosomes or sister chromatids fail to separate properly. The extra chromosome causes an imbalance in gene expression, leading to abnormal growth, developmental defects, and genetic disorders. The severity of the condition depends on the chromosome involved. Common examples include:

- Trisomy 21 (Down syndrome)
- Trisomy 18 (Edwards syndrome)
- Trisomy 13 (Patau syndrome)

15. Write release factor of translation and termination in prokaryotes and eukaryotes.

During translation termination, a release factor (RF) recognizes a stop codon (UAA, UAG, or UGA) in the A-site of the ribosome. It promotes the hydrolysis of the bond between the polypeptide chain and tRNA, releasing the newly synthesized protein and causing the ribosome to dissociate.

Prokaryotes	Eukaryotes
RF1 recognizes UAA and UAG stop codons.	eRF1 recognizes all three stop codons (UAA, UAG, and UGA).
RF2 recognizes UAA and UGA stop codons.	eRF3 is a GTP-binding protein that works with eRF1 to promote termination.
RF3 is a GTPase that helps release RF1 and RF2 from the ribosome.	eRF3 hydrolyzes GTP and assists in the release of the completed polypeptide chain and ribosome recycling.

16. Define epigenetic.

Epigenetics

Epigenetics is the study of heritable changes in gene expression that occur without changing the DNA sequence. These changes regulate whether a gene is turned on or off through chemical modifications such as DNA methylation, histone modification, and chromatin remodeling. Epigenetic changes play an important role in cell differentiation, growth, development, and response to environmental factors. Abnormal epigenetic modifications can contribute to various diseases, including cancer and genetic disorders.

17. Discuss hydrostatic pressure relationship of fluid depth.

Hydrostatic Pressure Relationship with Fluid Depth

Hydrostatic pressure is the pressure exerted by a fluid at rest due to the weight of the fluid above it. It increases with increasing depth because a greater column of fluid exerts more weight. The relationship is expressed by the equation:

$$P = \rho g h$$

Where:

- **P** = hydrostatic pressure
- **ρ** = density of the fluid
- **g** = acceleration due to gravity
- **h** = depth of the fluid

From this equation, hydrostatic pressure is directly proportional to the depth of the fluid. Therefore, the deeper an object is submerged, the greater the pressure acting on it.

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18. Enlist components of translation.

Components of Translation

The main components required for translation are:

- ✓ ***mRNA (Messenger RNA):*** Carries the genetic code from DNA to the ribosome.
- ✓ ***Ribosomes:*** The site of protein synthesis, composed of large and small subunits containing rRNA and proteins.
- ✓ ***tRNA (Transfer RNA):*** Transfers specific amino acids to the ribosome according to the mRNA codons.
- ✓ ***Amino acids:*** The building blocks used to synthesize proteins.
- ✓ ***Aminoacyl-tRNA synthetases:*** Enzymes that attach the correct amino acid to its corresponding tRNA.
- ✓ ***Translation factors:*** Initiation factors (IFs/eIFs), elongation factors (EFs/eEFs), and release factors (RFs/eRFs) that regulate the different stages of translation.
- ✓ ***Energy molecules (ATP and GTP):*** ATP is used for charging tRNA, while GTP provides energy for initiation, elongation, and termination of translation.

19. Differentiate between prokaryotic and eukaryotic mRNA.

Difference between Prokaryotic and Eukaryotic mRNA

Prokaryotic mRNA	Eukaryotic mRNA
Synthesized in the cytoplasm.	Synthesized in the nucleus.
Usually polycistronic, encoding multiple proteins.	Usually monocistronic, encoding a single protein.
No 5′ cap or 3′ poly(A) tail.	Contains a 5′ cap and 3′ poly(A) tail.
No introns; splicing is generally absent.	Contains introns, which are removed by RNA splicing.
Transcription and translation occur simultaneously.	Transcription and translation occur separately.
Has a short half-life and is degraded rapidly.	More stable because of mRNA processing and protective modifications.

20. What is chromosomal duplication?

Chromosomal Duplication

Chromosomal duplication is a type of chromosomal mutation in which a segment of a chromosome is copied and repeated, resulting in an extra copy of one or more genes. It usually occurs due to unequal crossing over during meiosis or errors in DNA replication. The duplicated chromosomal segment increases the amount of genetic material, which may alter gene expression and affect normal development. Chromosomal duplication can be beneficial, harmful, or neutral, depending on the genes

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21. What are splicing pathways?

Splicing Pathways

Splicing pathways are the mechanisms by which introns are removed from pre-mRNA and exons are joined together to produce mature mRNA in eukaryotic cells. Splicing is essential for the accurate expression of genes before translation. The major splicing pathways are:

- ✓ ***Spliceosome-mediated splicing:*** The most common pathway, carried out by the spliceosome, which contains snRNPs (U1, U2, U4, U5, and U6)**.** It removes introns and joins exons to form mature mRNA.
- ✓ ***Self-splicing:*** In this pathway, certain introns remove themselves without the help of the spliceosome. It occurs in Group I and Group II introns, which act as catalytic RNA (ribozymes).
- ✓ ***Alternative splicing:*** A process in which a single pre-mRNA is spliced in different ways to produce multiple mRNA variants and different proteins from the same gene. This increases protein diversity in eukaryotic cells.

22. Explain why lag time forms between DNA replication and methylation. How mismatch repair system use lag and its advantages?

After DNA replication in bacteria, the newly synthesized DNA strand is temporarily unmethylated, while the parental strand remains methylated. This creates a lag period between DNA replication and methylation because the enzyme DNA adenine methylase (Dam methylase) methylates the new strand only after replication is completed. This temporary difference allows the mismatch repair (MMR) system to distinguish the newly synthesized strand from the original parental strand. When a mismatch is detected, MutS recognizes the mismatch, MutL binds to MutS, and MutH identifies the unmethylated daughter strand at a nearby GATC site. MutH cuts the new strand, the incorrect DNA segment is removed, DNA polymerase synthesizes the correct sequence using the parental strand as a template, and DNA ligase seals the nick.

Advantages of the lag in methylation:

- It enables the mismatch repair system to identify the newly synthesized strand correctly.
- It ensures that replication errors are corrected accurately.
- It reduces the mutation rate and maintains genomic stability.
- It improves the fidelity of DNA replication and prevents the accumulation of harmful mutations.

23. Explain removal of introns.

Removal of Introns

The removal of introns is called RNA splicing, a process that occurs in the nucleus of eukaryotic cells after transcription. During splicing, the non-coding introns are removed from the pre-mRNA, and the coding exons are joined together to form mature mRNA. This process is carried out by the spliceosome, a large ribonucleoprotein complex composed of snRNPs (U1, U2, U4, U5, and U6) and associated proteins. The spliceosome recognizes specific sequences at the 5′ splice site, branch point, and 3′ splice site, cuts out the intron as a lariat (loop) structure, and joins the adjacent exons together. The mature mRNA is then transported from the nucleus to the cytoplasm, where it is translated into a protein. RNA splicing is essential for the accurate expression of genes and the production of functional proteins.

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24. Why chromosomal aberration do occurs in new born babies and first trimester abortion?

Chromosomal aberrations are changes in the number or structure of chromosomes. They most commonly occur due to errors during meiosis, especially nondisjunction, where chromosomes fail to separate properly during gamete formation. As a result, the embryo receives an abnormal number of chromosomes.

In newborn babies, chromosomal abnormalities that are compatible with life may lead to genetic disorders such as Down syndrome (Trisomy 21), Edwards's syndrome (Trisomy 18), and Patau syndrome (Trisomy 13). These abnormalities can cause developmental delays, congenital defects, and intellectual disabilities.

In first-trimester abortions (miscarriages), chromosomal aberrations are the most common cause. Severe abnormalities, such as monosomy, trisomy, or major structural chromosome defects, often prevent normal embryonic development. As a result, the pregnancy ends spontaneously during the first trimester because the embryo is not viable.

Therefore, chromosomal aberrations are a major cause of both congenital disorders in newborns and early pregnancy loss, depending on the type and severity of the chromosomal defect.

25. Differentiate between Autopolyploidy and Allopolyploidy.

Difference between Autopolyploidy and Allopolyploidy

Autopolyploidy	Allopolyploidy
Arises from the duplication of chromosome sets within the same species.	Arises from the combination of chromosome sets from two different species, followed by chromosome doubling.
All chromosome sets are genetically similar.	Chromosome sets are derived from different species.
Usually occurs due to failure of chromosome separation during cell division.	Usually occurs through hybridization between two species and subsequent chromosome doubling.
Homologous chromosomes may form multivalents during meiosis, often reducing fertility.	Chromosomes usually form bivalents, resulting in more stable meiosis and higher fertility.
Common in many cultivated plants such as potato and alfalfa.	Common in crops such as bread wheat, cotton, and tobacco.

26. What is non-disjunction? Enlist causes of non-disjunction.

Non-disjunction

Non-disjunction is the failure of homologous chromosomes or sister chromatids to separate properly during cell division (meiosis or mitosis). As a result, the daughter cells receive an abnormal number of chromosomes (aneuploidy). Non-disjunction is a major cause of chromosomal disorders such as Down syndrome, Turner syndrome, and Klinefelter syndrome. Causes of Non-disjunction are:

1. Errors during meiosis I or meiosis II, leading to improper chromosome separation.
2. Defects in the spindle fibers, preventing normal movement of chromosomes.
3. Failure of centromere division or improper attachment of chromosomes to spindle fibers.
4. Advanced maternal age, which increases the risk of chromosome segregation errors.
5. Genetic mutations affecting proteins involved in chromosome segregation.
6. Exposure to radiation, chemicals, or other environmental factors that interfere with cell division.

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27. Explain process of pinocytosis.

Process of Pinocytosis

Pinocytosis is a type of endocytosis in which a cell takes in extracellular fluid and dissolved substances through small membrane-bound vesicles. It is commonly known as "cell drinking." The process begins when the plasma membrane folds inward (invaginates) around the extracellular fluid. This invagination deepens and encloses the fluid along with dissolved molecules. The edges of the membrane then fuse, forming a small vesicle inside the cell. The vesicle separates from the plasma membrane and moves into the cytoplasm, where its contents are processed or transported to lysosomes for digestion. Pinocytosis is an energy-dependent (ATP-requiring) process and occurs continuously in most eukaryotic cells. It helps in the uptake of nutrients, ions, hormones, and other dissolved molecules, and plays an important role in maintaining normal cellular functions.

28. Write role of intermediate filament.

Role of Intermediate Filaments

Intermediate filaments are strong, rope-like fibers of the cytoskeleton with a diameter of about 10 nm. They provide mechanical strength and structural support to cells and help maintain their shape. The major roles of intermediate filaments are:

- Provide mechanical stability and maintain the structural integrity of cells.
- Resist stretching and mechanical stress, preventing cell damage.
- Anchor the nucleus and other organelles in their proper positions.
- Form the nuclear lamina, which supports the nuclear envelope.
- Strengthen cell-to-cell junctions such as desmosomes and hemidesmosomes.
- Help maintain the shape and organization of tissues, especially in epithelial, muscle, and nerve cells.

29. What is the metabolic pathway of PKU in the enzyme deficiency?

Metabolic Pathway of PKU (Phenylketonuria) in Enzyme Deficiency

Phenylketonuria (PKU) is an autosomal recessive metabolic disorder caused by a deficiency or absence of the enzyme phenylalanine hydroxylase (PAH). Normally, this enzyme converts the amino acid phenylalanine into tyrosine with the help of the cofactor tetrahydrobiopterin (BH_4). In PKU, the deficiency of phenylalanine hydroxylase blocks this metabolic pathway, causing phenylalanine to accumulate in the blood and tissues. The excess phenylalanine is converted into abnormal metabolites, such as phenylpyruvate, phenyllactate, and phenylacetate, which are excreted in the urine. The accumulation of phenylalanine is toxic to the developing brain and can lead to intellectual disability, developmental delay, seizures, and behavioral problems if left untreated. PKU is managed by a low-phenylalanine diet, supplementation with tyrosine, and in some patients, BH_4 therapy.

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30. Write role of formal methionine in bacteria.

Role of Formyl Methionine (fMet) in Bacteria

Formyl methionine (fMet) is the first amino acid incorporated during the initiation of protein synthesis in bacteria. It is a modified form of methionine in which a formyl group is attached to the amino group. The roles of formyl methionine are:

- It serves as the initiator amino acid for protein synthesis in prokaryotes.
- It is carried to the ribosome by the initiator tRNA ($tRNA^{fMet}$).
- It recognizes the start codon (AUG) and initiates translation.
- It helps the 30S ribosomal subunit correctly assemble the initiation complex.
- After protein synthesis begins, the formyl group, and often the methionine itself, may be removed by specific enzymes to produce the mature protein.

31. What are binding sites in ribosomes?

Binding Sites in Ribosomes

Ribosomes contain three binding sites that are involved in protein synthesis during translation. These sites bind tRNA molecules and ensure the correct addition of amino acids to the growing polypeptide chain.

1. ***A (Aminoacyl) Site:*** The A site is the entry site for the aminoacyl-tRNA carrying the next amino acid. It recognizes the mRNA codon and positions the correct tRNA.
2. ***P (Peptidyl) Site:*** The P site holds the tRNA carrying the growing polypeptide chain. Peptide bond formation occurs between the amino acid in the A site and the polypeptide attached to the tRNA in the P site.
3. ***E (Exit) Site:*** The E site is where the uncharged (empty) tRNA moves after transferring its amino acid. It then leaves the ribosome.

32. Explain stages of mitosis.

Stages of Mitosis

Mitosis is the process of nuclear division in which one parent cell produces two genetically identical daughter cells, each having the same number of chromosomes as the parent cell.

- ✓ ***Prophase:*** Chromatin condenses into visible chromosomes, each consisting of two sister chromatids joined at the centromere. The nucleolus disappears, the nuclear envelope begins to break down, and spindle fibers start to form.
- ✓ ***Metaphase:*** Chromosomes align at the equatorial plane (metaphase plate) of the cell. Spindle fibers attach to the centromeres through kinetochores, ensuring proper chromosome alignment.
- ✓ ***Anaphase:*** The centromeres divide, and the sister chromatids separate. They are pulled by spindle fibers toward opposite poles of the cell.
- ✓ ***Telophase:*** The separated chromosomes reach the opposite poles and begin to uncoil into chromatin. New nuclear envelopes and nucleoli reappear around each set of chromosomes, forming two daughter nuclei.
- ✓ ***Cytokinesis:*** After mitosis, the cytoplasm divides, producing two genetically identical daughter cells with the same chromosome number as the parent cell.

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33. Why double strand breakage is more difficult than single bond break in DNA?

A double-strand break (DSB) is more difficult to repair than a single-strand break (SSB) because both strands of the DNA molecule are broken, resulting in the loss of the intact complementary strand that normally serves as a template for repair. In contrast, during a single-strand break, the undamaged strand remains intact and guides accurate repair. Double-strand breaks require complex repair mechanisms such as homologous recombination (HR) or non-homologous end joining (NHEJ)**.** These processes are more complicated and, particularly in the case of NHEJ, can introduce mutations or chromosomal rearrangements if the DNA ends are joined incorrectly. If double-strand breaks are not repaired properly, they can cause genomic instability, chromosome loss, translocations, cell death, or cancer. Therefore, double-strand breaks are considered more severe and biologically dangerous than single-strand breaks.

34. Differentiate between monosomy and trisomy.

Monosomy	Trisomy
A chromosomal abnormality in which one chromosome is missing, resulting in 2n − 1 chromosomes.	A chromosomal abnormality in which one extra chromosome is present, resulting in 2n + 1 chromosomes.
Usually caused by nondisjunction during meiosis.	Also caused by nondisjunction during meiosis.
The affected individual has only one copy of a particular chromosome.	The affected individual has three copies of a particular chromosome.
Often causes severe developmental abnormalities and many cases are not compatible with life.	May cause developmental disorders, but some individuals survive depending on the chromosome involved.
Example**:** Turner syndrome **(**45, X**).**	***Examples*****:** Down syndrome (Trisomy 21), Edwards's syndrome (Trisomy 18), and Patau syndrome (Trisomy 13).

35. What is self-splicing?

Self-splicing is a process in which certain introns remove themselves from a pre-RNA molecule without the help of the spliceosome or protein enzymes. These introns act as ribozymes (catalytic RNA molecules) and catalyze their own excision. Self-splicing occurs mainly in Group I and Group II introns, which are found in some bacteria, lower eukaryotes, and the organelles (mitochondria and chloroplasts) of eukaryotic cells. After the intron is removed, the exons are joined together to produce a continuous RNA molecule. Self-splicing is important because it allows RNA processing without the spliceosome, demonstrating that RNA can function both as genetic material and as a biological catalyst.

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36. Mention events of prophase I in meiosis I.

Prophase I is the longest and most complex stage of Meiosis I. It is divided into five sub-stages, during which the following events occur:

- ✓ ***Leptotene:*** Chromosomes begin to condense and become visible as long, thin threads.
- ✓ ***Zygotene:*** Homologous chromosomes pair together in a process called synapsis, forming bivalents (tetrads)**.**
- ✓ ***Pachytene:*** Crossing over occurs between non-sister chromatids of homologous chromosomes, resulting in genetic recombination.
- ✓ ***Diplotene:*** The synaptonemal complex disappears, and homologous chromosomes begin to separate but remain attached at chiasmata, the sites of crossing over.
- ✓ ***Diakinesis:*** Chromosomes become highly condensed, chiasmata move toward the chromosome ends (terminalization), the nuclear membrane and nucleolus disappear, and spindle fibers begin to form, preparing the cell for metaphase I.

37. Why UVA is harmful to skin as compared to UVB?

UVA is considered more harmful to the skin than UVB because it has a longer wavelength (320–400 nm), allowing it to penetrate deeper into the dermis. This deep penetration damages collagen and elastin fibers, leading to premature skin aging (photoaging), wrinkles, and loss of skin elasticity. Although UVB (280–320 nm) has higher energy and mainly affects the epidermis, causing sunburn and direct DNA damage, UVA is present throughout the day, passes through clouds and window glass, and produces reactive oxygen species (ROS) that indirectly damage DNA. Continuous UVA exposure increases the risk of skin cancer, including melanoma, and contributes significantly to long-term skin damage. Therefore, UVA is considered more harmful for chronic skin damage and aging, while UVB is mainly responsible for sunburn and acute DNA damage.

38. What will happen if topoisomerase do not performs its function?

Topoisomerase is an essential enzyme that relieves the torsional stress and supercoiling generated during DNA replication and transcription. If topoisomerase does not perform its function, the DNA ahead of the replication fork becomes highly supercoiled, preventing the replication machinery from moving forward. As a result, DNA replication slows down or stops, and DNA strands may break due to excessive tension. This can lead to replication fork collapse, DNA damage, chromosome instability, and mutations. In severe cases, the cell cannot complete DNA replication, causing cell cycle arrest or cell death. Therefore, topoisomerase is essential for normal DNA replication, transcription, chromosome segregation, and maintenance of genomic stability.

39. What is fibrous protein? Explain its structure, properties and examples.

Fibrous proteins are long, thread-like proteins that mainly provide structural support, strength, and protection to cells and tissues. They are generally insoluble in water and play an important role in maintaining the body's framework.

Structure:

Fibrous proteins consist of long polypeptide chains arranged in parallel and organized into α-helices, β-sheets, or triple helices**.** The chains are held together by hydrogen bonds, disulfide bonds, and other intermolecular interactions, making them strong and stable. Unlike globular proteins, they have a repetitive amino acid sequence and an elongated shape.

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Properties:

- Long, narrow, and thread-like in shape.
- Insoluble or only slightly soluble in water**.**
- Provide mechanical strength and structural support.
- Highly stable and resistant to stretching and mechanical stress.
- Primarily perform structural rather than catalytic functions.

Examples:

- ***Collagen:*** The most abundant protein in connective tissues, bones, tendons, and skin.
- ***Keratin:*** Found in hair, nails, skin, feathers, and horns; provides protection and toughness.
- ***Elastin:*** Present in ligaments, lungs, and blood vessels; provides elasticity and flexibility.
- ***Fibroin:*** The main structural protein of silk produced by silkworms and spiders.

40. Name the checkpoint present in mitosis.

The major checkpoints associated with mitosis and the cell cycle are:

- ✓ ***G1 Checkpoint (Restriction Checkpoint):*** Ensures that the cell has adequate size, nutrients, growth factors, and undamaged DNA before entering the S phase.
- ✓ ***G2/M Checkpoint:*** Verifies that DNA replication is complete and that any DNA damage has been repaired before the cell enters mitosis.
- ✓ ***Spindle Assembly Checkpoint (Metaphase Checkpoint):*** This is the main checkpoint during mitosis. It ensures that all chromosomes are correctly attached to spindle fibers through their kinetochores before sister chromatids separate during anaphase. If any chromosome is improperly attached, mitosis is temporarily halted until the error is corrected.

41. Write a detailed note on Down's syndrome.

Down's syndrome

Down's syndrome is a chromosomal disorder caused by the presence of an extra copy of chromosome 21. It is also known as Trisomy 21 because affected individuals have three copies of chromosome 21 instead of two, resulting in a total of 47 chromosomes.

Cause:

The main cause of Down's syndrome is nondisjunction during meiosis, in which chromosome 21 fails to separate properly during the formation of gametes. As a result, an abnormal gamete containing an extra chromosome fuses with a normal gamete, producing an embryo with an additional chromosome 21.

Types of Down's syndrome:

1. ***Trisomy 21 (Nondisjunction type):*** The most common type, where all body cells contain an extra chromosome 21.
2. ***Translocation Down's Syndrome:*** A part of chromosome 21 becomes attached to another chromosome, usually chromosome 14 or 21.
3. ***Mosaic Down's Syndrome:*** Some cells contain the extra chromosome 21, while other cells have the normal chromosome number due to errors occurring after fertilization.

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Symptoms and Characteristics:

- Intellectual disability and delayed development.
- Characteristic facial features such as a flat facial profile and upward-slanting eyes.
- Short stature and reduced muscle tone (hypotonia).
- Delayed speech and motor development.
- Increased risk of congenital heart defects.
- Hearing and vision problems.
- Increased susceptibility to certain infections and health complications.

Diagnosis:

Down's syndrome can be diagnosed through:

- Prenatal screening tests such as ultrasound and maternal blood tests.
- Chromosomal analysis (karyotyping) to confirm the presence of an extra chromosome 21.
- Diagnostic tests such as amniocentesis and chorionic villus sampling (CVS).

42. Write role of msh2, msh3 and msh4.

Role of MSH2, MSH3, and MSH4

MSH2, MSH3, and MSH4 are members of the MutS family of proteins that play important roles in DNA repair and genetic recombination**.**

1. MSH2 (MutS Homolog 2):

- MSH2 is a key protein involved in the mismatch repair (MMR) system.
- It forms complexes with MSH6 or MSH3 to recognize and bind DNA mismatches and small insertion/deletion loops produced during DNA replication.
- It helps maintain genomic stability by preventing the accumulation of replication errors.
- Mutations in the MSH2 gene can increase the risk of Lynch syndrome and certain cancers.

2. MSH3 (MutS Homolog 3):

- MSH3 forms a complex with MSH2 called MutSβ (MSH2–MSH3 complex)**.**
- It mainly recognizes and repairs larger insertion/deletion loops caused by replication errors.
- It also participates in the repair of certain types of DNA damage and contributes to maintaining chromosome stability.

3. MSH4 (MutS Homolog 4):

- MSH4 functions mainly during meiosis rather than general mismatch repair.
- It forms a complex with MSH5 and helps in the proper formation of crossovers between homologous chromosomes**.**
- It promotes genetic recombination and ensures accurate chromosome segregation during meiosis.

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43. Write components, structure and functions of cytoskeleton.

Cytoskeleton

The cytoskeleton is a network of protein filaments present in the cytoplasm of eukaryotic cells that provides structural support, maintains cell shape, and helps in cellular movement and organization. It consists of three main components: microfilaments, intermediate filaments, and microtubules**.** Microfilaments are the thinnest fibers, about 7 nm in diameter, and are composed mainly of actin proteins. They help maintain cell shape, enable cell movement, and participate in muscle contraction. Intermediate filaments are approximately 10 nm in diameter and are composed of proteins such as keratin, vimentin, and lamins. They provide mechanical strength, maintain cellular stability, and support the nuclear envelope. Microtubules are the largest cytoskeletal structures, about 25 nm in diameter, and are made up of α-tubulin and β-tubulin. They form hollow tubular structures involved in chromosome movement, intracellular transport, and the formation of the mitotic spindle during cell division.

The cytoskeleton performs several important functions in the cell. It maintains the shape and structural integrity of cells, provides resistance against mechanical stress, and helps in the movement of cells and organelles. It assists in the transport of vesicles and other cellular components within the cytoplasm. During cell division, the cytoskeleton forms the spindle apparatus required for accurate chromosome separation. It also provides structural support to specialized cellular structures such as cilia and flagella, making it essential for proper cellular organization and function.

44. Differentiate between cytokinesis of plants and animals.

Difference between Cytokinesis of Plants and Animals

Cytokinesis in Plants	Cytokinesis in Animals
It occurs by the formation of a cell plate between the two daughter nuclei.	It occurs by the formation of a cleavage furrow that divides the cytoplasm.
The cell plate develops from Golgi-derived vesicles that fuse at the center of the cell.	The cleavage furrow is formed due to the contraction of actin and myosin filaments in the cell membrane.
Cytokinesis usually begins at the center and progresses toward the outer edges of the cell.	Cytokinesis begins at the cell membrane and proceeds inward toward the center**.**
A new cell wall and plasma membrane are formed between the daughter cells.	No new cell wall is formed; only the plasma membrane separates the daughter cells.
It is slower because of the presence of a rigid cell wall**.**	It is generally faster because animal cells lack a rigid cell wall.
Common in plant cells and other cells with cell walls**.**	Common in animal cells and other cells without cell walls**.**

45. How ozone layer plays its role in preventing DNA damage?

The ozone layer plays an important role in preventing DNA damage by absorbing most of the harmful ultraviolet (UV) radiation coming from the Sun, especially UV-C and a large portion of UV-B radiation. UV radiation can damage DNA by causing the formation of pyrimidine dimers, especially thymine dimers, which distort the DNA structure and interfere with DNA replication and transcription. The ozone layer acts as a protective barrier by reducing the amount of high-energy UV radiation reaching the Earth's surface, thereby decreasing DNA mutations and cellular damage. If the ozone layer becomes depleted, more UV-B radiation reaches living organisms, increasing the risk of DNA mutations, skin cancers, premature skin aging, and damage to immune cells.

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46. How many gametes are produced in male and female after meiosis?

After meiosis, the number of gametes produced differs in males and females due to differences in the processes of spermatogenesis and oogenesis.

In males (Spermatogenesis):

One primary spermatocyte undergoes meiosis and produces four haploid sperm cells (gametes). All four daughter cells develop into functional sperm.

In females (Oogenesis):

One primary oocyte undergoes meiosis and produces one functional haploid ovum (egg cell) and three polar bodies. The polar bodies usually degenerate and do not participate in fertilization.

47. What is threshold effect in multifactorial disorder?

Threshold Effect in Multifactorial Disorders

The threshold effect is a concept in multifactorial disorders where a disease appears only when the combined effect of multiple genetic and environmental factors exceeds a certain critical level (threshold).

In these disorders, many genes contribute small effects along with environmental factors such as diet, lifestyle, or exposure to harmful conditions. Individuals whose total risk remains below the threshold do not show symptoms, while those whose combined genetic and environmental risk crosses the threshold develop the disease.

Examples of multifactorial disorders showing a threshold effect include cleft lip and palate, neural tube defects, diabetes, hypertension, and congenital heart defects.

The threshold effect explains why a disease may occur in one individual but not in another, even among family members, because the total combination of genetic susceptibility and environmental influences differs between individuals.

48. Explain initiation and elongation in bacteria.

INITIATION

Initiation is the first stage of protein synthesis in bacteria, where the ribosome assembles on the mRNA. It begins when the 30S ribosomal subunit binds to the Shine-Dalgarno sequence on mRNA with the help of initiation factors (IF1 and IF3). The initiator tRNA carrying formyl methionine (fMet-tRNA^{fMet}) binds to the start codon AUG at the P site of the ribosome. Then, the 50S ribosomal subunit joins the complex with the help of IF2 and GTP, forming the complete 70S initiation complex. This prepares the ribosome for protein synthesis.

ELONGATION

Elongation is the stage in which amino acids are added continuously to form a growing polypeptide chain. During this process, an aminoacyl-tRNA carrying the next amino acid enters the A site with the help of elongation factors and GTP. A peptide bond is formed between the amino acids by the peptidyl transferase activity of the ribosome. The ribosome then moves along the mRNA in a process called translocation, shifting the tRNA from the A site to the P site and the empty tRNA to the E site for exit. This cycle continues until the complete polypeptide chain is synthesized.

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49. Write role of actin and myosin in cytokinesis.

Actin and myosin play a crucial role in cytokinesis of animal cells by forming the contractile ring beneath the plasma membrane at the equator of the dividing cell. Actin filaments form a ring around the middle of the cell, while myosin II interacts with these filaments and uses energy from ATP to slide them past one another. This interaction causes the contractile ring to tighten, creating a cleavage furrow that gradually constricts the cell membrane. As the cleavage furrow deepens, the cytoplasm is divided into two equal parts until the cell is completely separated into two genetically identical daughter cells. Thus, actin provides the structural framework, whereas myosin generates the force required for contraction, making both proteins essential for successful cytokinesis in animal cells.

50. Differentiate between non sense and frame shift mutation.

Difference between Nonsense Mutation and Frameshift Mutation

Nonsense Mutation	Frameshift Mutation
Caused by a single base substitution that changes a codon into a stop codon.	Caused by the insertion or deletion of one or more nucleotides that is not a multiple of three.
Results in premature termination of protein synthesis.	Changes the reading frame of the mRNA from the point of mutation onward.
Produces a shortened (truncated) protein, which is usually non-functional.	Produces a protein with an altered amino acid sequence, often resulting in a non-functional protein.
Only one codon is directly changed into a stop codon.	All downstream codons are altered due to the shift in the reading frame.
Example: Mutation changing UAU (Tyrosine) to UAA (Stop codon).	***Example:*** Insertion or deletion of a single nucleotide in a gene, causing a shift in the reading frame.

51. Differentiate between Aneuploidy and Euploidy.

Difference between Aneuploidy and Euploidy

Aneuploidy	Euploidy
A condition in which there is a gain or loss of one or a few individual chromosomes.	A condition in which the entire set of chromosomes is increased or decreased.
The chromosome number is not an exact multiple of the haploid number (e.g., 2n ± 1).	The chromosome number is an exact multiple of the haploid number (e.g., 3n, 4n, 6n).
Usually caused by nondisjunction during meiosis or mitosis.	Usually caused by failure of complete chromosome sets to separate or by chromosome doubling.
Commonly results in genetic disorders and developmental abnormalities.	Common in plants and may increase size, vigor, and adaptability.
Examples: Down syndrome (Trisomy 21), Turner syndrome (45, X), and Klinefelter syndrome (47, XXY).	***Examples:*** Triploid banana (3n), tetraploid potato (4n), and hexaploid wheat (6n).

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52. Write classification of chromosomal aberration.

Classification of Chromosomal Aberrations

Chromosomal aberrations are changes in the number or structure of chromosomes. They are divided into two main types**:**

1. Numerical Chromosomal Aberrations:

These occur when the number of chromosomes changes.

- ✓ ***Aneuploidy:*** Gain or loss of one or a few chromosomes.
- ✓ ***Monosomy (2n − 1):*** One chromosome is missing.
- ✓ ***Trisomy (2n + 1):*** One extra chromosome is present.
- ✓ ***Euploidy:*** Change in the whole set of chromosomes.
  - Triploidy (3n)
  - Tetraploidy (4n)

2. Structural Chromosomal Aberrations:

These occur when the structure of a chromosome changes due to breakage.

- ✓ ***Deletion:*** Loss of a chromosome segment.
- ✓ ***Duplication:*** Repetition of a chromosome segment.
- ✓ ***Inversion:*** A chromosome segment breaks and rejoins in the reverse direction.
- ✓ ***Translocation:*** A chromosome segment moves to another chromosome.

53. Which component is adequate to mismatch nick on 5' mismatch?

For a 5′ mismatch nick, the main component responsible for removing the mismatched DNA is the 5′→3′ exonuclease activity of DNA polymerase I (in bacteria). In the bacterial mismatch repair (MMR) system, MutS recognizes the mismatch, MutL recruits other repair proteins, and MutH makes a nick in the newly synthesized unmethylated DNA strand. When the nick is located on the 5′ side of the mismatch, DNA polymerase I (5′→3′ exonuclease) removes the incorrect DNA segment. DNA polymerase then fills the gap with the correct nucleotides, and DNA ligase seals the remaining nick. Thus, the 5′→3′ exonuclease activity of DNA polymerase I is the key component involved in repairing a 5′ mismatch nick.

54. What is proofreading mechanism of RNA polymerase?

RNA polymerase proofreading is the process by which RNA polymerase detects and corrects mistakes made during RNA synthesis (transcription). This mechanism helps improve the accuracy of the RNA transcript. When an incorrect nucleotide is added to the growing RNA chain, RNA polymerase pauses and moves backward (backtracking). It then removes the incorrect nucleotide using its intrinsic nuclease activity or with the help of proofreading factors (such as GreA and GreB in bacteria). After the wrong nucleotide is removed, RNA polymerase resumes transcription and adds the correct nucleotide. Although the proofreading ability of RNA polymerase is less efficient than the proofreading activity of DNA polymerase, it significantly reduces transcription errors and helps produce an accurate RNA molecule.

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55. What is carrier and channel protein?

Carrier Protein and Channel Protein

Carrier proteins bind to a specific molecule or ion, undergo a change in shape, and transport the substance across the membrane. They are highly specific and can transport substances by facilitated diffusion or active transport using energy (ATP). Examples include the glucose transporter (GLUT) and the sodium–potassium (Na^+/K^+) pump**.**

Channel proteins form hydrophilic pores or channels in the cell membrane through which specific ions or water molecules pass. They do not change shape for each transport cycle and allow rapid movement of substances down their concentration gradient by facilitated diffusion. Examples include potassium channels, sodium channels, calcium channels, and aquaporins**.**

56. What is gene duplication?

Gene duplication is a type of mutation in which an extra copy of a gene is produced in the genome. It usually occurs due to unequal crossing over during meiosis, errors in DNA replication, or chromosomal duplication. The duplicated gene provides an additional copy while the original gene continues to perform its normal function. Over time, the extra copy may accumulate mutations and develop a new function or increase the production of a particular protein. Therefore, gene duplication is an important source of genetic variation and evolution. Examples include the globin gene family (α- and β-globin genes) and the opsin genes responsible for color vision in humans.

57. Write a short note of P53 protein.

P53 Protein

P53 is a tumor suppressor protein, often called the "guardian of the genome", because it protects cells from genetic damage. It is produced by the TP53 gene and plays a vital role in maintaining genomic stability. When DNA is damaged, P53 stops the cell cycle to allow time for DNA repair. If the damage is successfully repaired, the cell continues to divide. However, if the damage is severe and cannot be repaired, P53 triggers apoptosis **(**programmed cell death**)** to prevent the damaged cell from becoming cancerous. Mutations in the TP53 gene cause loss of P53 function, allowing abnormal cells to divide uncontrollably. As a result, TP53 mutations are found in many human cancers, making P53 one of the most important proteins involved in cancer prevention.

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